CS 2340 Term Project User Manual

This guide assumes you have already installed the MARS IDE. See the README for instructions on installing MARS.

How to Set Up (MARS Options)

In order to assemble this program, **open** `src/driver.asm` in MARS, then go to *Settings* in the Menu bar and **ensure "Assemble all files in directory" is checked**, or the program will not assemble correctly. Then, you can assemble and run.

At this point, you can either follow the instructions printed to the standard output or the rest of this manual—they detail the same steps.

This program uses MARS' Bitmap display for the GUI, and the MMIO keyboard and display simulator to read input / display text output. Once you hit run, **open** the **Bitmap Display** tool and **Keyboard and Display MMIO Simulator** tool from the *Tools* menu on the menu bar. Then do the following:

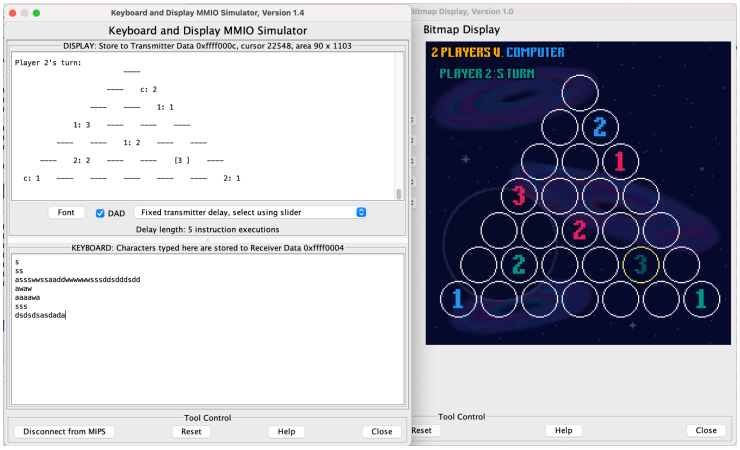
1) **Configure the Bitmap Display** with the following values (change **unit width to 2**, **unit height to 2**, and **display height to 512**)

Unit Width in Pixels:	2
Unit Height in Pixels:	2
Display Width in Pixels:	512
Display Height in Pixels:	512
Base address for display:	0x10010000 (static data)

2) Hit **Connect to MIPS** button **on the bottom left of Bitmap Display** (make sure you connect Bitmap Display FIRST)

3) Hit **Connect to MIPS** button **on bottom left of Keyboard & Display MMIO Simulator** (make sure you connect this AFTER Bitmap Display)

For best game experience (so you can see the game board GUI, keyboard input, and text display all at the same time), it's recommended you set up your windows like so:



How to Play

First select the game mode from the menu (1-player or 2-player) using the WASD (representing up, left, down, right, respectively) keys as arrow keys. You can hit RETURN, TAB, or SPACE to select a mode.

To play, similarly you can use the WASD keys to toggle between circles, and hit RETURN, TAB, or SPACE to finalize a spot you've selected. If you're having trouble toggling between spots using the up/down (W and S) keys, since they only move to immediate above and below neighbors and don't work if the neighbors already have numbers, the left and right (A and D) keys will always work to go to the previous or next free circle on the board, respectively.